INCH-POUND

MIL-C-24643/48C
14 March 1994
SUPERSEDING
MIL-C-24643/48B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSDNW
(INCLUDING VARIATION LSDNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSDNW (unarmored), and LSDNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated (see table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Cross-linked polyethylene insulation or ethylene propylene rubber insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth - Two conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fifth - Binder tape over each pair applied helically with overlap.

Construction variant for type LSDNW

* Sixth - Cross-linked polyolefin jacket (see table I for thickness).

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Construction variant for type LSDNWA

* Sixth - Cross-linked polyolefin jacket (see table I for thickness).
* Seventh - Braided metal armor

* TABLE I. Details.

Military 1/ part no. M24643/48	Size 2/	AWG conductor size	Insulation thickness min avg (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
					TYPE LSDNW		TYPE LSDNWA	
					min	max	min	max
-01UN	3	16 (class B)	0.030	0.040	0.361	0.390	0.411	0.440
-02UN	4	14 (class B)	.030	.040	.398	.430	.448	.480
-03UN	9	10 (class B)	.030	.040	.504	.545	.554	.595
-04UN	14	9 (class B)	.045	.040	.564	.610	.614	.660
-05UN	23	7 (class B)	.045	.045	.638	.690	.688	.740
-06UN	50	3 (class C)	.045	.050	.842	.910	.892	.960
-07UN	75	1 (class C)	.055	.055	1.000	1.080	1.050	1.130
-08UN	100	0 (class D)	.055	.055	1.082	1.170	1.132	1.220

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSDNW and LSDNWA.

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage Withstand (minimum) (volts)	Cold Bending mandrel diameter (max inch)	Accelerated service loading current (amperes)
3	16 (class B)	4.3	6000	4000	4	----
4	14 (class B)	2.68	5000	4000	4	----
9	10 (class B)	1.08	3800	4000	5	----
14	9 (class B)	.86	4400	5500	6	100
23	7 (class B)	.54	3700	5500	7	130
50	3 (class C)	.21	2600	5500	9	215
75	1 (class C)	.13	2600	7000	10	285
100	0 (class D)	.11	2400	7000	11	335

1/ Type LSDNW and LSDNWA.

EXAMINATION AND TESTS:

Requirements

Basic electrical:

Conductor Resistance - ohms/1000 feet at 25°C, max ..... (see table II)
Voltage Withstand - volts, root mean square, minimum ..... (see table II)
*Insulation resistance - Conductor to conductor
Megohms/1000 feet minimum ..... (see table II)
Conductor continuity ..... No failure

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EXAMINATION AND TESTS (continued):

	Requirements
Group A:	
Visual and dimensional	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air Oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50

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EXAMINATION AND TESTS (continued):

	Requirements
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - method 1 only, cycles, minimum	25
*Permanence of printing (jacket) - (type LSDNW only) cycles, minimum	125
*Armor type LSDNWA only - conformance to material, construction and coverage	No failure

Group D:

* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and Size	Feet (nominal)
LSDNW-3	2500
LSDNW-4 through 23	2000
LSDNW-50	1500
LSDNW-75, 100	1000

Note: Not for AIR FORCE use.

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